The Transformer Architecture

A concise guide to the model behind modern language AI

Overview

The Transformer, introduced in the 2017 paper *Attention Is All You Need* (Vaswani et al.), replaced recurrence and convolution with a mechanism called **attention**. Instead of reading a sequence one token at a time, a Transformer looks at every token at once and learns which other tokens matter most for each position. This makes training highly parallel on GPUs and lets the model capture long-range dependencies far more easily than RNNs or LSTMs.

The original design is an **encoder-decoder** model built for machine translation. The encoder turns the input sentence into a set of context-rich vectors; the decoder uses those vectors, plus the tokens it has already generated, to predict the next output token. Both halves are stacks of identical layers (N = 6 in the paper).

High-level architecture

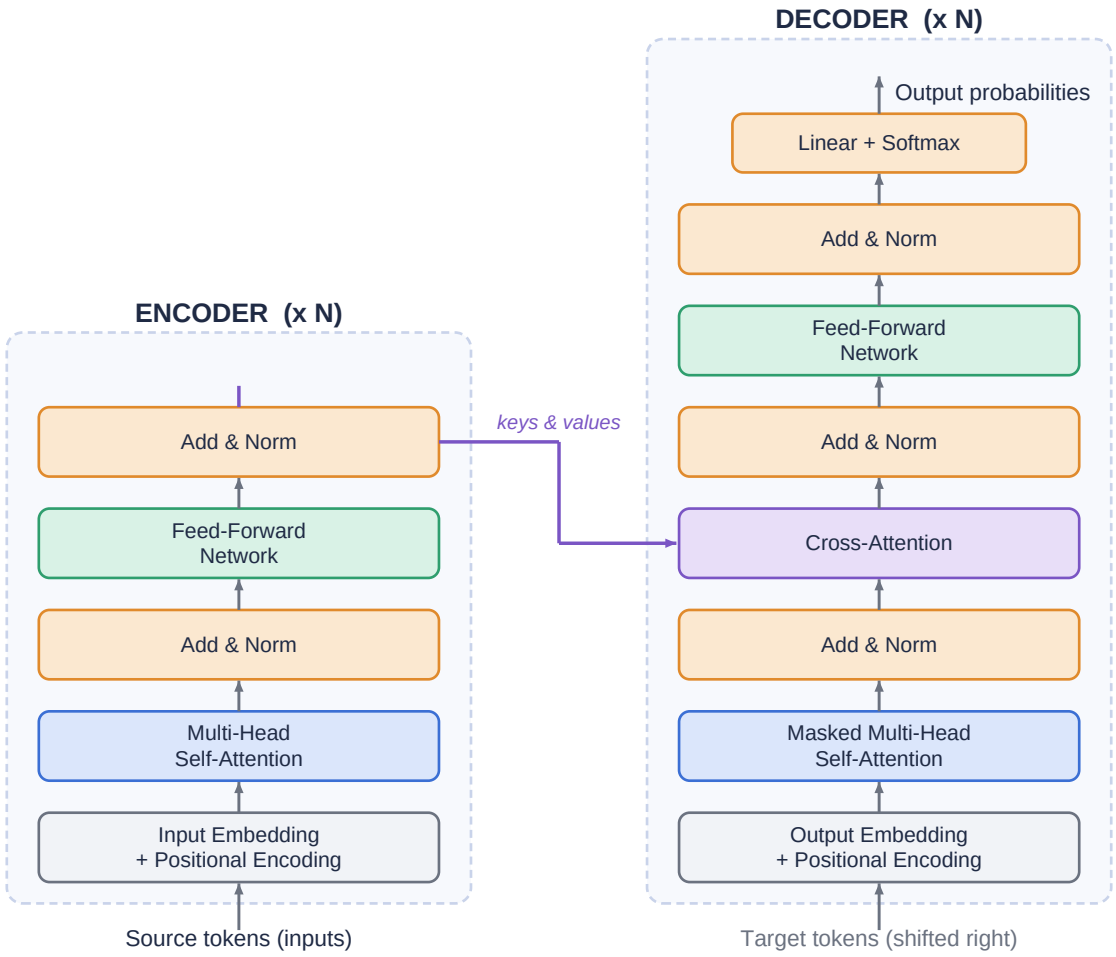

Figure 1. Encoder-decoder Transformer. Each layer wraps its sub-blocks with a residual connection and layer normalization (Add & Norm).

Key ideas at a glance

1. Self-attention mixes information between tokens. **2. Feed-forward networks** process each token independently. **3. Residual connections + layer norm** keep deep stacks stable. **4. Positional encodings** inject word-order information, because attention alone is order-blind.

Self-Attention: The Core Mechanism

Each input token embedding is projected into three vectors using learned weight matrices: a **Query** (what am I looking for?), a **Key** (what do I contain?), and a **Value** (what information do I pass on?). The similarity between a query and every key decides how much of each value flows into the output.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(Q K^{T} / \sqrt{d_k} \right) V$$

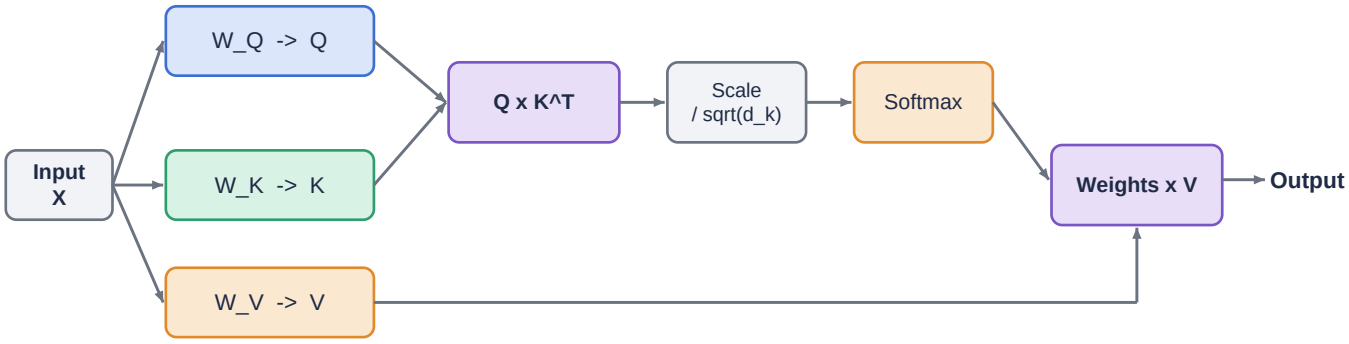

Figure 2. Scaled dot-product attention. Dividing by sqrt(d_k) keeps the dot products from growing too large, which would push softmax into regions with tiny gradients.

Reading an attention map

The softmax produces a matrix of weights in which each row sums to 1. A row shows how one token distributes its attention across the whole sentence. In the example below (illustrative values), the verb *sat* attends strongly to its subject *cat*.

Figure 3. Example attention weights (illustrative).

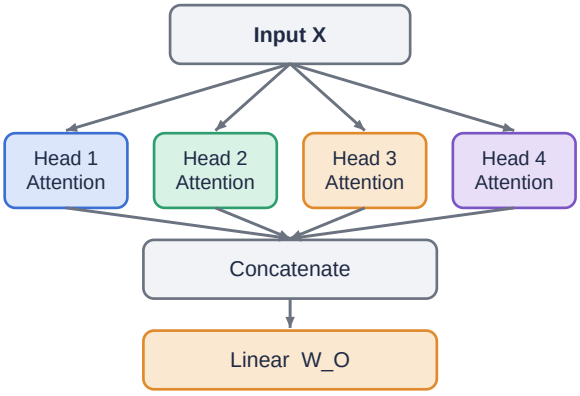

Figure 4. Multi-head attention: several heads run in parallel.

Multi-head attention

A single attention operation can focus on only one kind of relationship at a time. Multi-head attention runs **h** attention operations in parallel (8 in the original paper), each with its own projections of dimension d_{model}/h. Different heads tend to specialise: some track syntax, others coreference or nearby words. Their outputs are concatenated and mixed by a final linear layer W_O.

Masked attention in the decoder hides future tokens by setting their scores to minus infinity before the softmax, so each position can only see earlier positions. **Cross-attention** takes queries from the decoder but keys and values from the encoder output, linking the two halves.

Positional Encoding

Because attention treats its input as an unordered set, the model must be told where each token sits. The original Transformer adds sinusoidal vectors of different frequencies to the embeddings: PE(pos, 2i) = sin(pos / $10000^{2i/d}$) and PE(pos, 2i+1) = cos(pos / $10000^{2i/d}$). Low dimensions oscillate quickly and high dimensions slowly, giving every position a unique signature. Many modern models use learned or **rotary (RoPE)** position schemes instead.

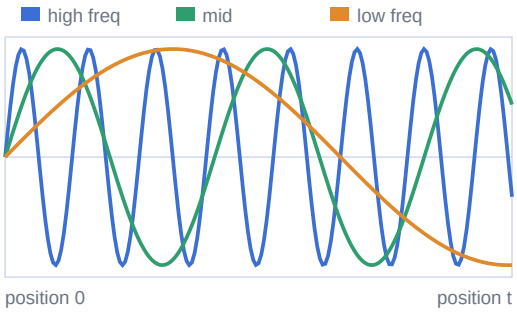

Figure 5. Sinusoids of different frequencies form a unique pattern per position.

Feed-Forward Network, Residuals and Normalization

After attention, every token passes through the same two-layer network: FFN(x) = max(0, $xW_1 + b_1)W_2 + b_2$. The hidden layer is typically 4x wider than the model dimension (2048 vs 512 in the paper) and is where much of the model's factual knowledge is thought to be stored. Each sub-layer is wrapped as **LayerNorm(x + Sublayer(x))**; the skip connection lets gradients flow through dozens of layers, and normalization stabilises training. Many newer models apply the norm *before* the sub-layer (pre-norm) for even steadier optimisation.

Variants and Applications

The same building blocks are rearranged into three major families:

Family	Structure	Examples	Typical use
Encoder-only	Bidirectional self-attention	BERT, RoBERTa	Classification, embeddings, search
Decoder-only	Causal (masked) attention	GPT family, Claude, Llama	Text generation, chat, code
Encoder-decoder	Encoder + cross-attention	T5, BART, original Transformer	Translation, summarisation

Beyond text, Transformers now power vision (ViT), speech recognition, protein structure prediction, and multimodal models. Their main cost is that attention scales quadratically with sequence length, which has driven work on FlashAttention, sparse and linear attention, and KV-caching for fast inference.

Reference: Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS.